

SOFTWARE ENGINEERING 381 - SEN381

CIVICCONNECT

Milestone 1

Project Engineering Document

Item	Details
Project	CivicConnect - Community Service Request Management Platform
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1. Source Basis and Contribution Boundary

This section is based on the SEN381 CivicConnect Master Project Brief and the Milestone 1 brief. The Master Brief defines CivicConnect as a community service request management platform and describes the current fragmented request process, the minimum requester/staff/management capabilities, and the mandatory project constraints (Belgium Campus ITiversity, 2026a).

No stakeholder interview evidence has been supplied at this stage. Therefore, stakeholder needs that come directly from the brief are treated as source-backed, while additional lifecycle stakeholders and influence/interest ratings are clearly identified as team analysis rather than interviewed fact.

2. Problem and Business Need

2.1 Problem Analysis

The organisation currently receives service requests through a fragmented combination of email, telephone calls, WhatsApp messages, spreadsheets and paper records. This creates recurring problems such as duplicate or lost requests, weak assignment and ownership, limited status visibility for requesters, inconsistent reporting, weak accountability for status/action changes, and inconsistent handling of sensitive information. The core engineering problem is therefore the absence of one controlled, traceable record of the service-request lifecycle rather than merely the absence of a new user interface (Belgium Campus ITiversity, 2026a).

Observed problem	Engineering implication
Requests may be duplicated, overlooked, incorrectly assigned or lost.	CivicConnect must support controlled request capture and clear ownership.
Requesters have limited visibility of request progress.	Requester-facing status, history and meaningful feedback are required business capabilities.
Staff struggle to prioritise, coordinate and identify ownership.	Authorised staff need useful request views, search/filter/sort and responsibility control.
Status/action accountability is weak.	The solution must preserve a controlled lifecycle and traceable actions.
Management information is unreliable or manual.	Structured request information must support oversight of open, overdue, resolved and closed work.
Sensitive information may be handled inconsistently.	Security and authorised access must be considered from the beginning of the lifecycle.

2.2 Business Need and Intended Value

The business need is for a controlled digital platform that provides a reliable, traceable and usable way to submit, manage, monitor and report service requests, while avoiding an unsustainable technical, operational or financial burden (Belgium Campus ITiversity, 2026a).

Value area	Expected value
Visibility	Requesters can determine whether their requests were received and what the current state is.
Accountability	Responsibility, actions and lifecycle changes are easier to trace.
Operational control	Staff can identify relevant work and ownership using consistent request information.

Value area	Expected value
Management oversight	Open, overdue, resolved and closed requests can be identified from controlled records.
Information quality	A single controlled lifecycle record reduces fragmented and inconsistent reporting.
Sustainability	Scope and later technology decisions remain constrained by team capacity, cost, quality, security and operational needs.

3. Stakeholder Analysis

The Master Brief explicitly describes three primary user groups: requesters, authorised staff, and management/oversight (Belgium Campus ITversity, 2026a). The analysis below also identifies project/lifecycle stakeholders that are implied by the organisation's business need and by mandatory operational/governance responsibilities. Those inferred stakeholders are labelled as such. Influence and interest ratings are team assessments for M1, not facts supplied by an interview.

ID	Stakeholder	Basis	Needs / expectations	Influence / interest	Main conflict / trade-off
STK-01	Requester	Explicit primary product stakeholder	Submit requests with appropriate information; categorise requests; view current status and history; receive meaningful feedback; have own information handled appropriately.	High / High	Transparency must not expose sensitive information to unauthorised people.
STK-02	Authorised service staff	Explicit primary product stakeholder	View relevant requests; search/filter/sort; view details; take/assign responsibility; update status through controlled transitions; record actions/comments; resolve/close where authorised.	High / High	Operational flexibility can conflict with lifecycle control and accountability.
STK-03	Management / oversight	Explicit primary product stakeholder	View useful service activity information; identify open/overdue/resolved/closed work; analyse by category/status or justified dimensions; support accountability and service-performance analysis.	High / High	Demand for broader reporting can expand scope and complexity.
STK-04	Organisation / service owner	Inferred business stakeholder	A sustainable controlled platform that improves visibility and accountability without excessive technical, operational or financial burden.	High / High	Low-cost expectations can conflict with resilience, security, monitoring or advanced functionality.
STK-05	System administration / operations	Inferred lifecycle stakeholder	Controlled configuration, access, secrets handling, deployment, logs, recovery and maintainable operation.	Medium / High	Operational controls consume effort that could otherwise be spent on visible features.
STK-06	SEN381 project team	Project delivery / engineering stakeholder	Clear scope, traceable requirements, manageable workload, controlled decisions, meaningful peer review and sufficient evidence to engineer and defend CivicConnect successfully.	High / High	Engineering governance and evidence activities consume schedule capacity that could otherwise be spent on implementation, but these controls are mandatory for project success.

The ratings above represent the team's M1 assessment rather than stakeholder interview evidence. Influence reflects the stakeholder's ability to affect product requirements, approval, operational use or project delivery, while interest reflects the extent to which CivicConnect's outcomes affect that stakeholder. These ratings will be reviewed if further stakeholder evidence becomes available.

3.1 Stakeholder Conflicts and M1 Responses

Conflict	Why it matters	M1 response
Requester transparency vs confidentiality	Requesters need meaningful status/history, but sensitive details must not be exposed to other requesters or unauthorised staff.	Baseline authorised access and controlled user-facing feedback; detailed identity design remains a later decision.
Staff flexibility vs lifecycle control	Staff must progress work efficiently, but unrestricted status changes would recreate weak accountability.	Require controlled status transitions and traceable actions; the exact transition catalogue remains an open business rule.
Management reporting vs scope discipline	Management may want many dashboards and reporting dimensions.	Commit only to the minimum oversight capabilities in the brief; advanced analytics/export remains deferred unless justified.
Low cost vs operational readiness	Free/low-cost services are preferred, but security, recovery and observability still matter.	Treat cost as a decision criterion, not permission to ignore quality/security/operations.
Feature breadth vs quality/evidence	Additional features consume time needed for requirements, reviews, testing, security and evidence.	Use a conservative scope baseline and assess additions through controlled change.

4. Scope Baseline

The Master Brief provides minimum business capabilities but states that these are not complete detailed requirements. The scope below converts those capabilities into a controlled M1 boundary and deliberately avoids committing the team to additional integrations or technology choices before evidence exists (Belgium Campus ITversity, 2026a; Belgium Campus ITversity, 2026b).

4.1 In Scope

- Direct submission of service requests into CivicConnect with appropriate request information.
- Selection of a request category through a controlled category mechanism.
- Requester access to the current status and history/list of previously submitted requests.
- Meaningful requester feedback when a request is accepted, rejected, updated or completed.
- Authorised staff access to relevant service requests, including search, filtering, sorting and full request details.
- Assignment or acceptance of responsibility for a request.
- Controlled request status updates and recording of relevant actions, comments or resolution information.
- Authorised resolution and closure of service requests.
- Management/oversight views that identify open, overdue, resolved and closed work and support category/status analysis.

4.2 Out of Scope for the Initial Baseline

- Automatic ingestion or synchronisation of requests from email, telephone calls, WhatsApp, spreadsheets or paper records. These channels describe the current problem, but direct integration is not a minimum capability.
- Payments, inventory, asset-management or unrelated business modules outside the service-request problem.
- AI chatbot or autonomous AI classification/decision-making. These are not minimum capabilities and would create additional accuracy, privacy, security and verification obligations.

4.3 Deferred / Future Scope

- External email/SMS/WhatsApp notifications, subject to later cost, privacy, integration and operational evidence.
- File/photo attachments, subject to storage, privacy, malware-scanning and cost analysis.
- Advanced analytics, data exports, automated escalation and advanced service-level management are deferred until detailed management rules and stakeholder value are validated. A basic rule for determining whether a request is overdue remains within the baseline because overdue request identification is a minimum management capability.
- Third-party integrations and single sign-on after organisational systems and technology/environment constraints are known.
- Delivery platform implementation, including whether CivicConnect will use a web-based, native mobile or other client interface, is deferred until evidence-based technology and interface decisions are made in a later milestone.

4.4 Deliberate Exclusion to Defend

Automatic legacy-channel integration is deliberately excluded from the initial baseline.

The current fragmented process uses email, telephone, WhatsApp, spreadsheets and paper, but the minimum CivicConnect capabilities require the platform to support request submission and lifecycle management; they do not require automatic ingestion from those channels. Including such integrations now would introduce third-party dependencies, duplicate-handling rules, privacy/security concerns, additional testing and schedule risk. A direct CivicConnect submission path therefore provides the required business capability while keeping the initial scope controllable. The team may reconsider integrations later through formal impact analysis if stakeholder value justifies them.

5. Assumptions and Constraints

5.1 Explicit Project Constraints and Mandatory Engineering Controls

The following project constraints and mandatory engineering controls are derived directly from the Master Project Brief (Belgium Campus ITiversity, 2026a). Their engineering implications represent the team's analysis of how these controls should influence later project work.

ID	Type	Constraint	Source	Engineering implication
C-001	Team size	Exactly 3 students.	Master Brief Section 4	All controlled work must fit a three-person team; substantive PR review capacity must be planned.
C-002	Schedule	Four formal milestones within the SEN381 delivery period.	Master Brief Section 4	Scope and evidence must progress incrementally; late reconstruction of evidence is not acceptable.
C-003	Cost / resources	Prefer free or low-cost services where practical and identify limits/likely operational cost.	Master Brief Section 4, Section 18	Later platform choices must consider total cost and free-tier limitations.
C-004	Scope	Committed scope must be baselined and controlled.	Master Brief Section 4, Section 14	Additional features must undergo impact analysis instead of being silently absorbed.
C-005	Quality	Quality attributes must be defined and later supported by measurable evidence.	Master Brief Section 4, Section 15	Requirements must include testable quality expectations linked to later verification.

ID	Type	Constraint	Source	Engineering implication
C-006	Security	Security is a lifecycle-wide responsibility, not a final add-on.	Master Brief Section 4, Section 16	Security/privacy concerns must influence requirements, repository controls, design, testing and deployment.
C-007	Technology	No stack, architecture or platform is prescribed.	Master Brief Section 4, Section 18.1	Technology selection must be deferred until evidence-based comparison against requirements, team capability, cost, security and deployment constraints.
C-008	Institutional environment	BC cannot guarantee support/compatibility for every technology or external service.	Master Brief Section 25	The team must investigate compatibility before committing and manage the risk of unavailable/unsupported technology.
C-009	Configuration management	Protected main, PRs and two non-author approvals are mandatory for substantive changes.	Master Brief Section 9	Review capacity becomes part of the schedule; substantive direct development on main is not acceptable.
C-010	AI accountability	AI output is not authoritative evidence; material use must be recorded and verified.	Master Brief Section 10	AI-assisted work must be independently checked and remains the team's responsibility.

5.2 Controlled Assumptions / Evidence Gaps

The following assumptions identify information that is not specified in the project brief and will require validation as further project evidence becomes available.

ID	Assumption / evidence gap	Treatment
A-001	The supplied project briefs are the current authoritative source for M1; no stakeholder interview evidence has been supplied at this stage.	Validate if the lecturer/client provides additional evidence.
A-002	CivicConnect is treated as a single-organisation platform; multi-tenant operation is not baselined.	Revisit only if a stakeholder requires multi-tenancy.
A-003	Users will have controlled identities/roles, but the identity mechanism is not chosen in M1.	Require evidence before the M2 authentication decision.
A-004	The brief does not provide production workload volumes.	Do not claim workload/performance facts until evidence exists.
A-005	The exact category catalogue, status-transition catalogue and overdue rule are not supplied.	Treat as high-priority open business-rule gaps. The team must define and validate these rules before detailed design; however, the requirement to identify overdue requests remains baselined.
A-006	The brief states that sensitive request information may exist but does not define detailed data-classification or retention rules.	Perform security/privacy analysis before persistence design.

5.3 Constraint Interactions / Ripple Effects

Interaction	Engineering consequence
Three-person team + two non-author approvals + fixed milestone schedule	Because every substantive PR needs review by both non-author members, review capacity is a real schedule constraint. Large late PRs would create a bottleneck. The team should therefore use smaller coherent changes and review progressively.
Low cost + quality/security/operational readiness	A free platform is not automatically acceptable. Future technology/platform decisions must still support secure configuration, testing, deployment, recovery and operational evidence.
Scope + schedule + quality	Adding optional features increases implementation and verification work. Under schedule pressure the correct response is controlled scope change, not silently reducing testing, security or evidence.

6. Requirements Analysis

The requirements below convert the CivicConnect minimum business capabilities, stakeholder needs and M1 constraints into a controlled requirements baseline. Each requirement has a stable identifier, source, priority and acceptance criterion so that it can be traced and verified as the project progresses (Belgium Campus ITversity, 2026a; Belgium Campus ITversity, 2026b).

6.1 Functional Requirements

ID	Source / Stakeholder	Priority	Requirement	Acceptance Criteria
FR-001	STK-01; Master Brief Section 3	High	CivicConnect shall allow a requester to submit a new service request with the required request information.	AC-FR-001: A request with valid mandatory information is saved with a unique request identifier and confirmation is shown; missing mandatory information prevents submission and produces validation feedback.
FR-002	STK-01; Master Brief Section 3	High	CivicConnect shall allow a requester to categorise a request using a controlled category mechanism.	AC-FR-002: A requester can select a category from the approved category catalogue and the selected category is stored with the request. The final catalogue remains subject to A-005 validation.
FR-003	STK-01; Master Brief Section 3	High	CivicConnect shall allow an authorised requester to view the current status and history/list of previously submitted requests.	AC-FR-003: An authorised requester can open their request list/history and see the current status for each submitted request.
FR-004	STK-01; Master Brief Section 3	High	CivicConnect shall provide meaningful requester feedback when a request is accepted, rejected, updated or completed.	AC-FR-004: For each tested accepted, rejected, updated or completed event, the authorised requester can view the resulting state, timestamp and requester-facing feedback.
FR-005	STK-02; Master Brief Section 3	High	CivicConnect shall allow authorised service staff to view relevant requests, search/filter/sort them using controlled criteria, and view full request details.	AC-FR-005: Staff can open request details, search by request ID, and filter/sort by category, status, owner and submission date. These provisional team-derived criteria require stakeholder validation before M2.
FR-006	STK-02; Master Brief Section 3	High	CivicConnect shall allow authorised staff to assign or accept responsibility for a request.	AC-FR-006: An authorised staff member can assign or accept responsibility and the current request owner is recorded and visible to authorised staff.
FR-007	STK-02; Master Brief Section 3; A-005	High	CivicConnect shall allow authorised staff to update request status through controlled transitions and record relevant actions, comments and resolution information.	AC-FR-007: Only transitions permitted by the approved status-transition catalogue are accepted; an allowed transition is recorded together with the related action/comment/resolution information.
FR-008	STK-02; Master Brief Section 3	High	CivicConnect shall allow authorised staff to resolve or close requests where permitted.	AC-FR-008: An authorised staff member can resolve/close an eligible request and the resulting lifecycle state and resolution information are recorded.
FR-009	STK-03; Master Brief Section 3	High	CivicConnect shall provide management with service-activity information that identifies open, overdue, resolved and closed requests and supports category/status analysis.	AC-FR-009: Management can identify open, overdue, resolved and closed requests and view request information grouped or filtered by category and status. The exact overdue rule remains subject to A-005 validation.
FR-010	STK-03; STK-04; Master Brief Sections 2-3	High	CivicConnect shall preserve enough request lifecycle information to support accountability and service-performance analysis.	AC-FR-010: For tested requests, management can identify ownership/status/resolution actions, actor and timestamps, including submission and resolution/closure times, supporting accountability and basic lifecycle-duration analysis.

6.2 Non-Functional Requirements

ID	Quality Attribute / Source	Priority	Requirement	Acceptance Criteria
NFR-001	Usability; STK-01/02/03; C-005	High	The core CivicConnect requester and staff workflows shall be usable without requiring external training material for the defined acceptance scenarios.	AC-NFR-001: All baselined core-task scenarios are completed without external training material or facilitator assistance; completion and observed errors are recorded as evidence.
NFR-002	Authorisation / privacy; C-006; A-006	High	CivicConnect shall restrict sensitive request information and privileged functions to authorised roles.	AC-NFR-002: All defined unauthorised-access test cases are denied, while authorised role test cases can access only the functions and information permitted to that role.
NFR-003	Secrets / configuration security; C-006; C-009	High	Credentials, API keys, tokens, private keys and other secrets shall not be committed to the source repository and shall be supplied through approved configuration mechanisms.	AC-NFR-003: Repository review finds no committed secrets/credentials, and any required secret is obtained through the approved environment/configuration mechanism in the test/deployment setup.
NFR-004	Auditability / accountability; STK-03; C-005	High	Important lifecycle changes shall be traceable for accountability.	AC-NFR-004: For every tested ownership, status and resolution change, the system records the actor, action and timestamp and makes the record available to an authorised user.

6.3 Requirement Baseline Notes

- The exact category catalogue, status-transition catalogue and overdue rule remain controlled business-rule gaps under A-005 and must be validated before detailed design.
- The identity/authentication mechanism is intentionally not selected in M1; the access-control requirements are baselined without prematurely selecting the implementation mechanism.
- No production workload volumes are available under A-004, so no unsupported production performance or scalability target is claimed in this baseline.
- Any change to a baselined requirement must be impact-analysed and reflected in the requirement, acceptance criteria, RTM, risks and affected downstream artefacts.
- Priority rationale: All committed M1 requirements are High because the FRs implement minimum business capabilities and the NFRs cover essential quality, security and accountability controls; optional functions remain deferred.
- FR-005 note: Its search/filter/sort criteria are provisional team-derived details for testability and require stakeholder validation before M2.

7. Initial Requirements Traceability Matrix

The RTM establishes the M1 traceability path from stakeholder/source to requirement and acceptance criteria, while reserving explicit fields for later design, implementation, test and release evidence (Belgium Campus ITversity, 2026b).

Source / Stakeholder	Requirement ID	Requirement	Priority	Acceptance Criteria	Change / Issue / PR	Design / Architecture	Implementation	Test	Acceptance / Release Evidence
STK-01 / Master Section 3	FR-001	Submit service request	High	AC-FR-001	Change: N/A; exact Issue/PR in live RTM.	Deferred to M2	Deferred to M3	Deferred to M3	Deferred to M4
STK-01 / Master Section 3	FR-002	Controlled categorisation	High	AC-FR-002	Change: N/A; exact Issue/PR in live RTM.	Deferred to M2	Deferred to M3	Deferred to M3	Deferred to M4
STK-01 / Master Section 3	FR-003	View status and history	High	AC-FR-003	Change: N/A; exact Issue/PR in live RTM.	Deferred to M2	Deferred to M3	Deferred to M3	Deferred to M4
STK-01 / Master Section 3	FR-004	Requester lifecycle feedback	High	AC-FR-004	Change: N/A; exact Issue/PR in live RTM.	Deferred to M2	Deferred to M3	Deferred to M3	Deferred to M4
STK-02 / Master Section 3	FR-005	Staff view/search/filter/details	High	AC-FR-005	Change: N/A; exact Issue/PR in live RTM.	Deferred to M2	Deferred to M3	Deferred to M3	Deferred to M4
STK-02 / Master Section 3	FR-006	Assign/accept responsibility	High	AC-FR-006	Change: N/A; exact Issue/PR in live RTM.	Deferred to M2	Deferred to M3	Deferred to M3	Deferred to M4
STK-02 / A-005	FR-007	Controlled status/actions/comments	High	AC-FR-007	Change: N/A; exact Issue/PR in live RTM.	Deferred to M2	Deferred to M3	Deferred to M3	Deferred to M4
STK-02 / Master Section 3	FR-008	Resolve/close requests	High	AC-FR-008	Change: N/A; exact Issue/PR in live RTM.	Deferred to M2	Deferred to M3	Deferred to M3	Deferred to M4
STK-03 / Master Section 3	FR-009	Management activity oversight	High	AC-FR-009	Change: N/A; exact Issue/PR in live RTM.	Deferred to M2	Deferred to M3	Deferred to M3	Deferred to M4
STK-03/04 / Master Sections 2-3	FR-010	Lifecycle accountability information	High	AC-FR-010	Change: N/A; exact Issue/PR in live RTM.	Deferred to M2	Deferred to M3	Deferred to M3	Deferred to M4
C-005 / STK-01/02/03	NFR-001	Core workflow usability	High	AC-NFR-001	Change: N/A; exact Issue/PR in live RTM.	Deferred to M2	Deferred to M3	Deferred to M3	Deferred to M4
C-006 / A-006	NFR-002	Role-based access/privacy	High	AC-NFR-002	Change: N/A; exact Issue/PR in live RTM.	Deferred to M2	Deferred to M3	Deferred to M3	Deferred to M4
C-006 / C-009	NFR-003	Secrets/configuration security	High	AC-NFR-003	Change: N/A; exact Issue/PR in live RTM.	Deferred to M2	Deferred to M3	Deferred to M3	Deferred to M4
C-005 / STK-03	NFR-004	Auditability of lifecycle changes	High	AC-NFR-004	Change: N/A; exact Issue/PR in live RTM.	Deferred to M2	Deferred to M3	Deferred to M3	Deferred to M4

The repository-hosted RTM is the live traceability record. Exact Issue/PR identifiers must be current there and shown as live evidence. Example M1 trace: STK-01 / Master Brief Section 3 -> FR-001 -> AC-FR-001 -> PED Section 6.1 / controlled requirements artefact -> live Issue/PR. Approved post-baseline changes will receive a Change ID and update the affected RTM, risks and downstream artefacts.

8. Initial Risk Register

Team scoring convention: Probability 1-5 x Impact 1-5 = Risk Score. Priority bands: 1-4 Low, 5-9 Medium, 10-14 High, 15-25 Critical.

ID	Risk	Cause	Probability	Impact	Priority	Mitigation	Contingency	Owner	Status
RISK-001	Requirements may be ambiguous or interpreted differently, causing rework or incorrect downstream decisions.	The brief contains broad capabilities and some business rules remain open.	3 - Possible	4 - Major	12 High	Use stable IDs, explicit sources, priorities and measurable acceptance criteria; team-review requirements and maintain the RTM.	Clarify the source; revise the requirement/AC through controlled change; update RTM and dependent artefacts.	L. Masia	Open
RISK-002	Scope may expand beyond the capacity of a three-person team and fixed milestone schedule.	Optional features/new requests may be accepted without adequate impact analysis.	4 - Likely	5 - Severe	20 Critical	Maintain explicit in/out/deferred scope and analyse schedule, resource, quality, security and risk impact before additions.	Defer lower-priority functionality; record the approved scope change and update requirements, RTM, risks and planning.	A. Nkayi	Open

ID	Risk	Cause	Probability	Impact	Priority	Mitigation	Contingency	Owner	Status
RISK-003	Uneven availability or contribution may delay artefacts, reviews or milestone readiness.	Academic workload, timetable clashes, absence and two-reviewer approval requirements may create bottlenecks.	3 - Possible	4 - Major	12 High	Use clear issue ownership/internal deadlines, small progressive PRs and regular progress checks; raise blockers early.	Redistribute work by team agreement, update ownership/deadlines and escalate persistent participation issues where necessary.	T. Ndlozi	Open
RISK-004	Security/privacy requirements may be identified too late, causing redesign or inappropriate information handling.	Roles, permissions, data classification and detailed security rules are not fully defined in M1.	3 - Possible	5 - Severe	15 Critical	Capture security/privacy requirements early; identify role/access expectations and sensitive data; review security impact in requirements and PRs.	Raise controlled change; revise affected requirements/data-access expectations, update traceability/risks and reconsider later architecture decisions.	T. Ndlozi	Open
RISK-005	A later technology stack/platform may be incompatible, unsupported or too costly.	Stack/platform selection is deferred and environment, team-skill and operating-cost evidence is incomplete.	3 - Possible	4 - Major	12 High	Keep selection deferred in M1; in M2 compare viable options against requirements, capability, environment, security, deployment and cost evidence.	Select the best viable alternative using recorded decision criteria; update ADR/decision record, schedule implications and risks.	T. Ndlozi	Open
RISK-006	Important requirements may later be difficult or expensive to verify.	Acceptance criteria or NFRs may use vague/subjective wording.	3 - Possible	4 - Major	12 High	Use testable FRs and measurable NFRs with acceptance criteria; review wording for ambiguity/testability.	Refine before baseline or process through controlled change after baseline; update RTM and later verification plans.	L. Masia	Open
RISK-007	The team may apply GitHub governance inconsistently, weakening traceability and review evidence.	Members may be unfamiliar with the workflow or treat document review controls as unnecessary overhead.	2 - Unlikely	4 - Major	8 Medium	Protect main; use issues, branches, PRs, two non-author approvals, templates, meaningful commits and resolved review conversations.	Create corrective issue, restore the controlled process, re-review affected work and record the corrective action.	T. Ndlozi	Monitoring
RISK-008	AI-assisted work may introduce incorrect assumptions, unsupported claims or misleading content.	Generative AI can produce plausible but inaccurate output that may be accepted without sufficient verification.	3 - Possible	4 - Major	12 High	Record material AI use; verify important claims against source/project evidence; require human review; never expose secrets/sensitive information.	Reject/revise incorrect output; correct affected artefacts, update AI/traceability records and re-review before baseline/merge.	T. Ndlozi	Monitoring

9. Forward Engineering Considerations

These considerations identify later lifecycle concerns that already influence M1 requirements, constraints, risks or assumptions. They do not represent final architecture, technology, database, deployment or implementation decisions.

ID	Forward Engineering Consideration	Why it matters now	Later decision / activity affected	Information still missing	Risk if ignored
FEC-001	Security, Authentication & Role-Based Access	Requester, staff and management have different capabilities, so access expectations already affect requirements and acceptance criteria.	Authentication, authorisation, architecture, API access control, security testing and deployment configuration.	Exact roles, permission boundaries, authentication mechanism, sensitive-data classification and detailed security requirements.	Inappropriate access may be enabled, causing redesign or security weaknesses.
FEC-002	Testability & Verification Strategy	M1 FRs, NFRs and acceptance criteria need to be verifiable later.	Test strategy, automation, integration/system/performance testing and release quality gates.	Final quality thresholds, test environments/data, performance expectations and testing tools.	Important requirements may prove vague, expensive or difficult to verify.
FEC-003	Deployment Environment & Platform Compatibility	The team should not select a stack without evidence that it can be developed, hosted and operated in available environments.	Technology-stack selection, architecture, CI/CD, hosting, configuration and production deployment.	Supported runtimes/platforms, hosting options, environment limitations, service availability, free-tier limits and operating costs.	A preferred stack may later be unavailable, incompatible, difficult to deploy or too expensive.
FEC-004	Data Architecture, Audit History & Recovery	CivicConnect must store requests, statuses, ownership, comments/actions and enough history for accountability/reporting.	Database design, API contracts, validation, backup/recovery, reporting and migration.	Detailed entities, retention, audit-history rules, backup frequency, recovery expectations and likely volume.	Poor assumptions may cause missing audit data, schema rework, unreliable reporting or weak recovery.
FEC-005	Observability, Logging & Operational Support	A system that works in development must still be diagnosable and supportable when deployed.	Logging, monitoring, health checks, deployment design, incident response and operational support.	Hosting platform, monitoring tools, support responsibilities, alerting and availability expectations.	Failures may be discovered only after users complain and incidents may be difficult to diagnose.
FEC-006	Maintainability, Changeability & Technical Debt	Requirements/workflows may evolve as stakeholder understanding improves or changes are requested.	Architecture, modularity, technology selection, API/data design, testing and maintenance.	Likely future changes, support expectations, architectural constraints and final technology choices.	Short-term choices may create unnecessary coupling/technical debt and make later changes expensive.

9.1 Initial Deployment and Operational Considerations

- No deployment platform is selected in M1. Later selection must be justified against requirements, team capability, environment compatibility, security, likely operational cost and vendor/platform limitations.
- Development, test, staging and production environments must be distinguished later; success in a development environment will not be treated as production-readiness evidence.

- Environment-specific configuration and secrets, deployment/release approval, rollback/recovery, logging, monitoring, health checks and failure detection must be considered before release.
- The deployment decision must revisit A-004 workload uncertainty, A-006 sensitive-data gaps, C-003 cost/resource limits and C-008 institutional environment compatibility.

10. Engineering Decision Log

Only genuine M1 decisions and deliberate deferments are recorded. Decisions that require architecture, technology or implementation evidence are intentionally deferred rather than guessed.

ID	Context	Constraints	Alternatives	Decision / Deferment	Rationale	Trade-offs	Risks	Evidence	Later consequence
DEC-001	Legacy request channels are part of the current problem context.	Scope, schedule, security/privacy, three-person capacity.	Automatically integrate email/phone/Whats App/spreadsheets/ paper; or require direct CivicConnect submission.	Exclude automatic legacy-channel ingestion from the initial baseline.	Direct submission satisfies the minimum capability without introducing unnecessary integrations.	Less convenience for legacy-channel users; lower integration/testing complexity.	Future stakeholder demand may require integration.	Section 4.4; STK-01/02; C-004.	Any later integration requires formal impact analysis and traceability updates.
DEC-002	The brief prescribes no stack, architecture or delivery platform.	C-003 cost, C-007 technology freedom, C-008 environment, A-004 workload evidence gap.	Select a familiar stack now; select a platform now; or defer pending evidence.	Defer final architecture, technology stack and delivery-platform choice to M2.	M1 lacks sufficient evidence for a defensible consequential technology decision.	Preserves options but postpones implementation certainty.	Late discovery of incompatibility /learning-curve risk.	Sections 4.3, 5.1; RISK-005; FEC-003.	M2 must compare viable alternatives using evidence and record the final decision/ADR.
DEC-003	Management must identify overdue requests, but advanced SLA/escalation rules are not defined.	Minimum management capability, A-005 open overdue rule, schedule/scope control.	Defer all overdue logic; implement advanced SLA/escalation now; or baseline only basic overdue identification.	Keep basic overdue identification in scope and defer advanced SLA/escalation management.	The brief explicitly requires overdue identification, while detailed escalation/SLA rules lack evidence.	A basic overdue rule still requires validation; avoids premature advanced workflow.	Poorly defined rule could create incorrect overdue classification.	Section 4.3; FR-009; A-005.	Validate the overdue rule before detailed design and update FR-009/RTM if it changes.
DEC-004	Several optional capabilities may add value but are not minimum M1 commitments.	Scope, schedule, cost, security, risk and three-person capacity.	Include notifications, attachments, advanced analytics, SSO/integrations now; or defer.	Defer external notifications, attachments, advanced analytics/exports, SSO and third-party integrations.	A smaller controlled baseline provides stronger engineering evidence than broad unfinished scope.	Optional user convenience is postponed; verification burden is reduced.	Deferred features may later become stakeholder priorities.	Section 4.3; RISK-002; C-004.	Any addition must pass impact analysis and update scope, requirements, RTM and risks.

11. GitHub and Team Governance

GitHub is used as the controlled engineering evidence environment for CivicConnect. The controls below reflect the repository governance established for M1 and the mandatory Master Project Brief requirements (Belgium Campus ITversity, 2026a).

Project / process approach. CivicConnect uses a milestone-driven iterative approach: meaningful work is tracked through issues/tasks, completed on task branches and integrated through Pull Requests with two non-author approvals. The team prefers small progressive changes, reviews the PED/RTM/risks/decisions at each milestone, and applies impact analysis to post-baseline changes before implementation.

Control	M1 baseline	Controlled evidence
Repository	One controlled CivicConnect team repository.	Team GitHub repository.
Main branch	Protected; substantive direct development on main is not permitted.	Repository ruleset / branch protection.
Work initiation	Meaningful substantive work is represented by an issue/task and performed on a task branch from current main.	.github/ISSUE_TEMPLATE/engineering-task.md; repository Issues.
Pull Requests	Substantive changes enter main through a Pull Request linked to the relevant issue where practical.	.github/pull_request_template.md; PR history.
Approvals	Two approvals are required from the two team members other than the author; self-approval is not accepted.	Ruleset and PR review history.
Review quality	Reviews must be meaningful; comments are addressed and conversations resolved before merge.	Review comments, corrections, re-reviews and approvals.
History	Commits, PRs, reviews and merges are progressive and authentic; bulk reconstruction is avoided.	Repository commit/PR history.
Secrets	Passwords, API keys, tokens, private keys and confidential credentials must not be committed.	Repository review and configuration practice.
Team Working Agreement	Task ownership, communication, review, conflict/delay handling, AI accountability and baseline approval are controlled in the team agreement.	docs/governance/team-working-agreement.md.

12. AI Usage Register

The live AI Usage Register is maintained as a controlled repository artefact at docs/governance/ai-usage-register.md. It is not duplicated in full inside the PED so that one current source of AI-accountability evidence is maintained.

At baseline review, the team must confirm that every material AI-assisted contribution is recorded with: Date, Student, Tool, Engineering task, AI contribution, Verification, Decision, and Issues found. AI output is not treated as authoritative evidence; the responsible student remains accountable for verification, understanding and defence of the result (Belgium Campus ITversity, 2026a; Belgium Campus ITversity, 2026b).

- The register must include the material AI assistance used for requirements/RTM work, stakeholder/scope/constraints work and any governance or other engineering artefact where AI materially contributed.
- Verification must describe what the student checked, changed, rejected or accepted rather than merely stating that AI was used.
- No credentials, secrets or inappropriate sensitive information may be supplied to AI tools.

13. Baseline Review and Sign-off

13.1 Version History

Version	Date	Status / change	Review / approval
0.1	8 September 2026	Initial integrated M1 working document.	In progress
0.9	9 September 2026	Complete M1 baseline candidate with requirements, RTM, risk, forward considerations, decision log, governance and sign-off structure integrated.	Ready for team baseline review
1.0	9 September 2026	Revised M1 baseline candidate after quality review; requirements, traceability and project approach strengthened.	Pending three-member sign-off

13.2 Baseline Readiness Check

Check	Status before sign-off
Problem, business need, stakeholder analysis, scope and constraints reviewed	Ready for team confirmation/sign-off
Functional/NFR requirements and acceptance criteria reviewed for clarity, priority, source and testability	Ready for team confirmation/sign-off
RTM checked for source -> requirement -> acceptance criteria and future evidence structure	Ready for team confirmation/sign-off
Risk Register and Forward Engineering Considerations reviewed	Ready for team confirmation/sign-off
Engineering Decision Log reviewed; deferred decisions have clear evidence needs	Ready for team confirmation/sign-off
GitHub governance and Team Working Agreement evidence checked	Ready for team confirmation/sign-off
AI Usage Register current and material AI use verified	Ready for team confirmation/sign-off
PED integrated, coherent and ready to baseline	Ready for team confirmation/sign-off

13.3 Team Sign-off

Baseline type: Milestone 1 - Engineering Foundation & Requirements Baseline. Target baseline version: PED v1.0.

Team member	Student number	Signature / approval	Date
Thaban Ndlozi	602341	T.Ndlozi	09/09/2026
Anele Nkayi	577168	A.Nkayi	09/09/2026
Lebogang Masia	601562	L. Masia	09/09/2026

14. References

Belgium Campus ITversity (2026a) SEN381 CivicConnect Master Project Brief: Integrated Team Software Engineering Project. Unpublished module project brief. Pretoria: Belgium Campus ITversity.

Belgium Campus ITversity (2026b) SEN381 CivicConnect Project Milestone 1: Engineering Foundation & Requirements Baseline. Unpublished milestone brief. Pretoria: Belgium Campus ITversity.